

Ali Ejaz Khan

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EDUCATION

North Allegheny Senior High School

High School Junior and Rising Senior

Pittsburgh, PA

Expected Graduation: 2027

RELEVANT COURSEWORK

Current Courses: Honors Linear Algebra, AP Computer Science, AP US History, AP Physics 1 & 2, AP Economics: Macro and Micro

Previous Courses: AP Calculus BC, AP Computer Science Principles, AP Human Geography, AP Statistics

HONORS AND AWARDS

American Mathematics Competition | *AMC 12B: 111; AIME: 7*

- One of three students in a grade of approximately 700 to qualify for the American Invitational Mathematics Examination.

MathCON | *National Finalist*

- Placed among approximately the top 600 competitors nationally.

CMIMC | *9th Place*

- Placed 9th in a coding and mathematical problem-solving competition hosted by Carnegie Mellon University.

USA Computing Olympiad | *Silver Division*

- Advanced to the Silver Division in a national algorithmic programming competition.

DECA | *2x District Champion, 2x State Champion, 2x ICDC Qualifier*

- Won state titles and qualified twice for the International Career Development Conference.

Pittsburgh Regional Science and Engineering Fair | *3rd Place*

- Earned 3rd place in Health and Medicine for research on biological regulators underlying skin aging.

Pittsburgh Junior Academy of Science | *1st Place, State and Regional*

- Earned first-place awards in Biology for presenting skin-aging research.

Competitive Cybersecurity | *5th at LYKNCTF; 18th at V1T CTF*

- Competed in global capture-the-flag events involving reverse engineering, cryptography, web exploitation, and forensics.

TESTING RESULTS

SAT: 1530, 730 Reading/Writing, 800 Math

PSAT: 1480/1520, 720/760 Reading/Writing, 760/760 Math

AP Scores: 5 on AP Calculus BC, 5 AP AB Subscore, 5 on AP Statistics, 5 on AP Computer Science Principles, 5 on AP Human Geography, 5 on AP Physics 1, 5 on AP Physics 2, 5 on AP CSA, 5 on AP Micro, 5 on Macro, 4 on APUSH

TECHNICAL SKILLS

Languages and Tools: R, RStudio, Python, PyTorch, Java, LaTeX, Overleaf

Quantitative Skills: Monte Carlo simulation, simulation and modeling, statistical analysis, mathematical application, data organization

Professional Skills: Communication, problem solving, leadership, mentoring, tutoring, presentation development

EXPERIENCE

Paid Intern

2025 – Present

Gibson Capital LLC

Pittsburgh, PA

- Developed an R-based Monte Carlo retirement simulator with 14 adjustable inputs and outputs including probability of success, ending portfolio values, and spending-floor analysis.
- Built charts and summary tables used to communicate retirement projections at a firm managing more than \$2.5B in client assets.
- Contributing to a financial paper based on the simulation model.

Researcher

2025 – Present

University of Pittsburgh

Pittsburgh, PA

- Analyzed single-cell RNA-seq data from young and aged human skin using clustering, pathway enrichment, and transcription-factor activity analysis.
- Contributed to transcriptomics software that enables researchers to perform detailed cell-level analysis.

Researcher

2026 – Present

Rutgers University

Remote

- Studying differential privacy and private algorithms with a Rutgers professor, including continual counting, binary-tree mechanisms, and privacy loss.
- Exploring how mathematical privacy guarantees can protect sensitive data during computation and machine-learning model training.

LEADERSHIP AND ACTIVITIES

Participant

Summer 2026

Stanford University Mathematics Camp (SUMaC)

Stanford University

- Selected for an intensive proof-based mathematics program focused on advanced problem solving and collaborative exploration.
- Developing a project on Hamming cubes, binary strings, distance, and combinatorial structure.
- 1 of 104 students chosen around the globe. 5 percent acceptance with an admission exam.

Chapter Vice President

2024 – Present

DECA

North Allegheny

- Lead member education and competition preparation for a chapter of more than 250 students.
- Created a mentoring and communication system connecting newer members with experienced international qualifiers.
- Mentor students in finance events, financial literacy, and professional presentation skills.

Financial Manager; Former Tutor and Student

2022 – Present

Steel City Codes

Pittsburgh, PA

- Began as an eighth-grade student before returning from ninth through eleventh grade to tutor younger students in introductory Java.
- Joining the executive team to support financial management and the continued growth of computer-science education programs.

Financial Manager

2023 – Present

Project Water

Pittsburgh, PA

- Support financial coordination, outreach, and fundraising for student-led clean-water and healthcare initiatives.
- Contributed to efforts that raised more than \$300,000 and supported two wells in Africa and two hospitals in Ghana.

Volunteer

2024 – Present

Concordia of Franklin Park

Franklin Park, PA

- Completed 400+ volunteer hours supporting resident activities, meal service, and everyday technology use.
- Developed an anti-fraud guide and helped residents identify suspicious emails, calls, and impersonation attempts.